

D1 Measurement: the three equations

1. Hallucination rate

$$H = \frac{|C_{\text{hal}}|}{|C_{\text{hal}}| + |C_{\text{sup}}|}$$

2. Provenance label

$$\text{label}(i) = \begin{cases} \text{CORRECT_IN_DATA} & c_{\text{true}} > 0 \wedge c_{\text{false}} = 0 \\ \text{WRONG_IN_DATA} & c_{\text{true}} = 0 \wedge c_{\text{false}} > 0 \\ \text{NOT_IN_DATA} & c_{\text{true}} = 0 \wedge c_{\text{false}} = 0 \\ \text{MIXED} & c_{\text{true}} > 0 \wedge c_{\text{false}} > 0 \end{cases}$$

3. Cross tabulation

$$P(\text{cause} = k \mid \text{mode} = j) = \frac{n_{jk}}{n_j}$$